Day 12 – Virtual Try-On Improvement & Learning Report

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Overview

Today, I focused on improving the existing 2D Virtual Try-On prototype developed earlier. The main goal was to understand how clothing alignment, pose detection, garment fitting and image preprocessing affect the overall try-on experience. Along with implementation work, I also explored different technologies, datasets and approaches that can be used to improve Virtual Try-On systems in the future.

Improvements Made to the 2D Virtual Try-On System

I continued working on the existing Virtual Try-On prototype and experimented with different methods to improve the output quality.

The first improvement was background removal from garment images. Earlier, clothing images contained unnecessary background portions which affected the overlay quality. By generating transparent garment images, the clothing could be placed more naturally on the user image.

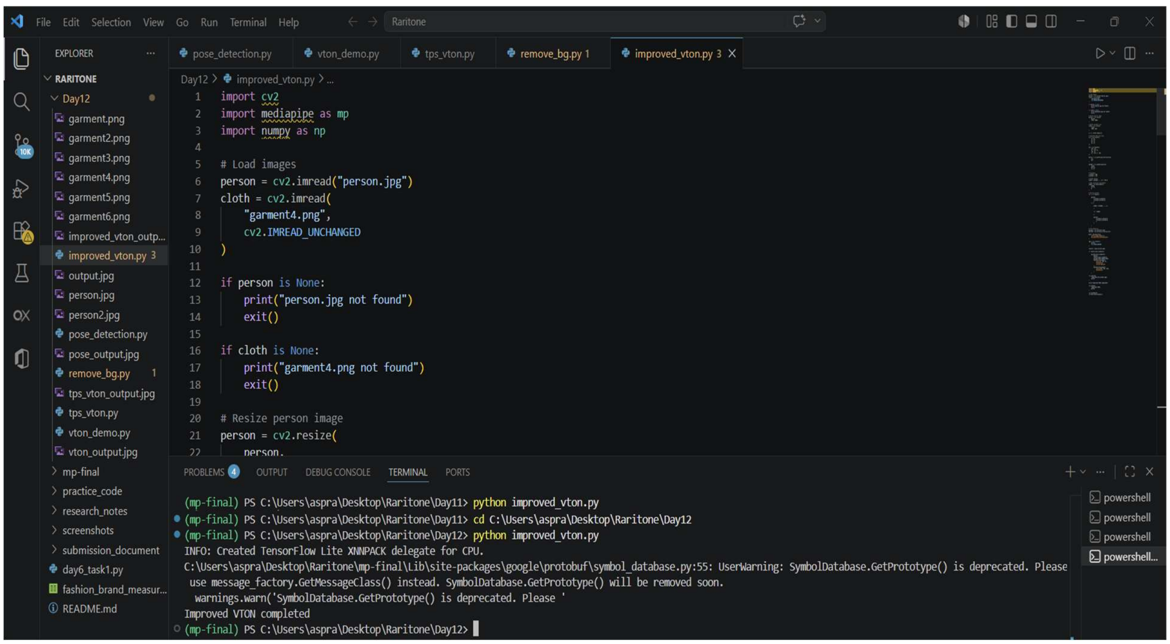
I also explored TPS-style garment transformation techniques to understand how garments can be adjusted before being placed on the user image. This helped me learn how image warping can improve garment positioning and fitting.

I tested different garment sizes, placements and overlay configurations to observe how alignment changes affect the final output. Through multiple experiments, I gained a better understanding of the challenges involved in making a garment appear naturally worn by a person.

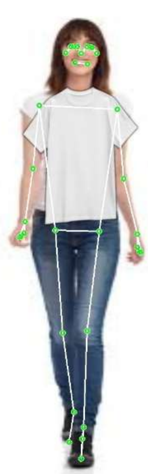
In addition, I continued using MediaPipe Pose for body landmark detection and observed how pose information can be used to guide garment placement.

Key Learnings

- Importance of accurate body landmark detection
- Effect of garment positioning on visual realism
- Role of transparent garment images in improving output quality
- Basics of garment warping and transformation
- Challenges involved in realistic virtual clothing fitting



Output:



3D Virtual Try-On Exploration

Apart from improving the 2D system, I spent time exploring how 3D Virtual Try-On systems work.

I learned about concepts such as:

- 3D avatar generation
- Body mesh creation
- Cloth simulation
- Avatar movement tracking
- Real-time rendering

I also explored tools commonly used in this domain, including Blender, Unity, Three.js and SMPL body models. This helped me understand the difference between simple image overlay approaches and full 3D clothing simulation systems.

Key Learnings

- How 3D avatars are generated
- Importance of body measurements in 3D fitting
- Basics of cloth simulation
- Challenges involved in real-time rendering
- Future possibilities for realistic Virtual Try-On applications

AI Model Exploration

During today's work, I explored different models that are commonly used in advanced Virtual Try-On systems.

- **MediaPipe Pose** was used in the current implementation for body landmark detection.
- **OpenPose** was studied to understand how detailed body keypoints can improve garment alignment.
- **Detectron2** was explored for its human segmentation capabilities and potential use in body-aware garment fitting.

- **U2Net** was reviewed as a segmentation model that can assist in background removal and foreground extraction tasks.

This exploration helped me understand how advanced Virtual Try-On systems combine pose estimation, segmentation and garment transformation techniques to generate more realistic results.

While MediaPipe was used in the current prototype, I researched how OpenPose and Detectron2 can provide more detailed body information and segmentation capabilities. I also explored how U2Net is commonly used for image segmentation and background removal tasks.

Key Learnings

- Different approaches to human pose estimation
- Importance of segmentation in clothing fitting
- Trade-offs between lightweight and advanced models
- Areas where model optimization can improve system performance

Dataset Exploration and Organization

I spent time researching datasets commonly used in fashion and Virtual Try-On projects.

Some datasets explored include:

- DeepFashion2
- VITON-HD

Key Learnings

- Importance of diverse fashion datasets
- Need for multiple body types and poses
- Dataset organization best practices
- Role of properly structured data in model development

Workflow Understanding

To better understand the complete system, I documented the current workflow of the Virtual Try-On pipeline.

The workflow consists of:

Input Image
↓
Image Preprocessing
↓
Pose Detection
↓
Garment Processing
↓
Background Removal
↓
Garment Transformation
↓
Overlay Generation
↓
Virtual Try-On Output

Documenting the workflow helped me visualize the complete process and identify areas where future improvements can be made.

Challenges Faced

During experimentation, I observed several practical challenges:

- Garment alignment was not always accurate
- Different image sizes affected fitting quality
- Overlay realism depended heavily on garment placement

- Background artifacts impacted visual appearance
- Creating realistic clothing fitting is more complex than simple image overlay

What I Learned

- Learned how pose detection helps in garment placement and alignment.
- Understood the importance of background removal for cleaner overlays.
- Explored TPS-style garment warping and its role in improving fitting quality.
- Gained insight into segmentation, human parsing and advanced Virtual Try-On techniques.
- Learned the basic concepts behind 3D avatar generation and cloth simulation.

Future Scope

- Improve garment fitting using body segmentation techniques.
- Integrate advanced pose estimation and segmentation models.
- Support multiple clothing categories and body poses.
- Explore 3D avatar-based Virtual Try-On systems.
- Enhance realism through cloth simulation and real-time rendering.

Conclusion

Today's work helped me gain a deeper understanding of Virtual Try-On systems and the technologies behind them. I improved the existing 2D prototype, explored garment transformation techniques, researched 3D try-on concepts, studied advanced pose and segmentation models, and reviewed relevant fashion datasets. Overall, the day provided valuable insights into both the implementation challenges and future possibilities of Virtual Try-On technology.